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Effects of Electrode Defects, Gas Adsorption, Electrolyte, and Temperature Variation on Capacitance and Charge Distribution in Crumpled Graphene EDLC Supercapacitors: A Molecular Dynamics Study

Presenters:

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What is EDLC Supercapacitor

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EDLC : Electrochemical double layer capacitors

- ✓Store charge electrostatically without chemical reaction
- ✓No charge transfer between the electrode and the electrolyte
- ✓Charge accumulates on the electrode surfaces because of the potential difference

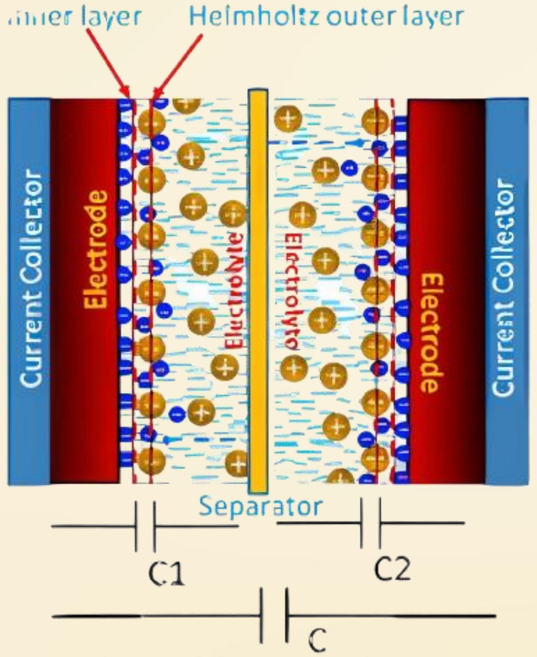
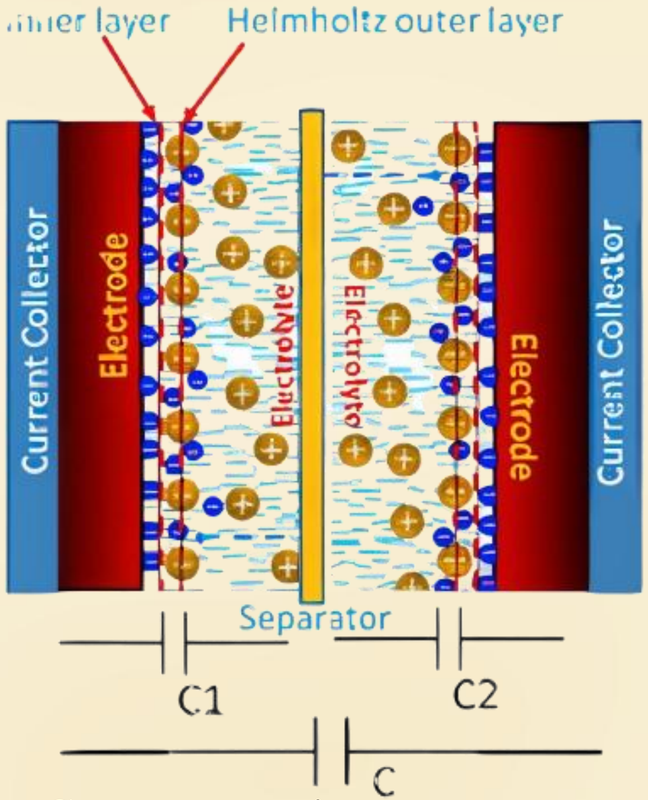


Fig. EDLC Supercapacitor

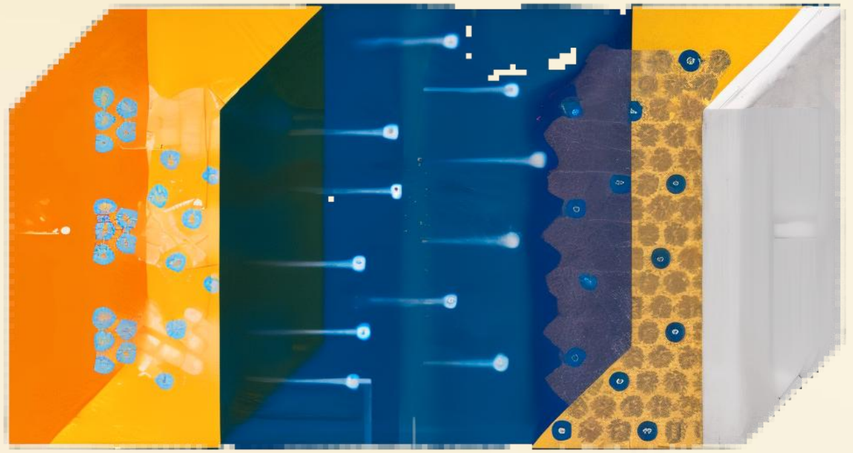
Why Supercapacitor is Better than Battery?

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Supercapacitor

- ✓High Power density
- ✓Low Energy density
- ✓High Life Cycle



Battery

- ✓Low Power density
- ✓High Energy density
- ✓Less Life Cycle

Why Supercapacitor is Better than Battery?

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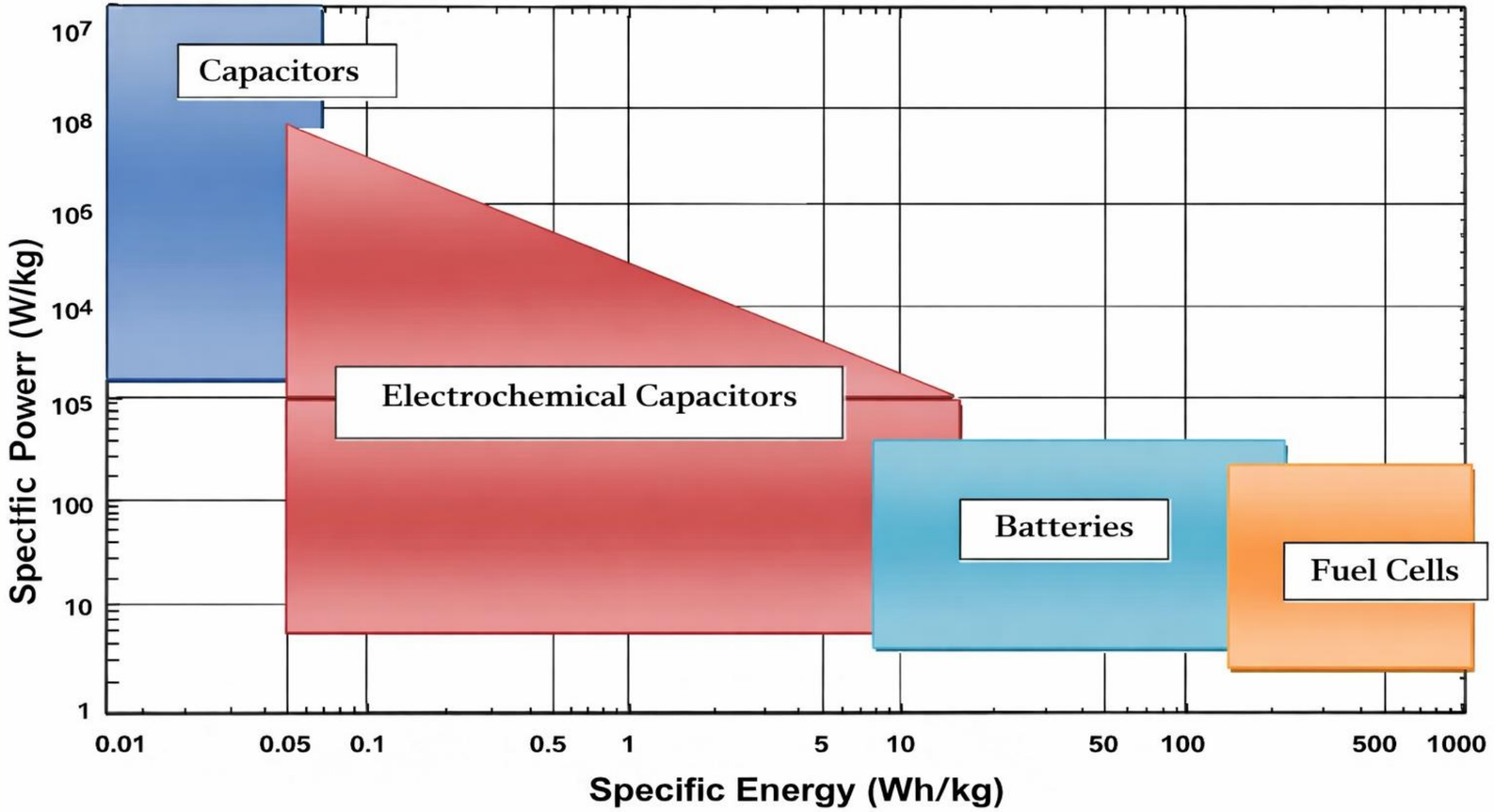
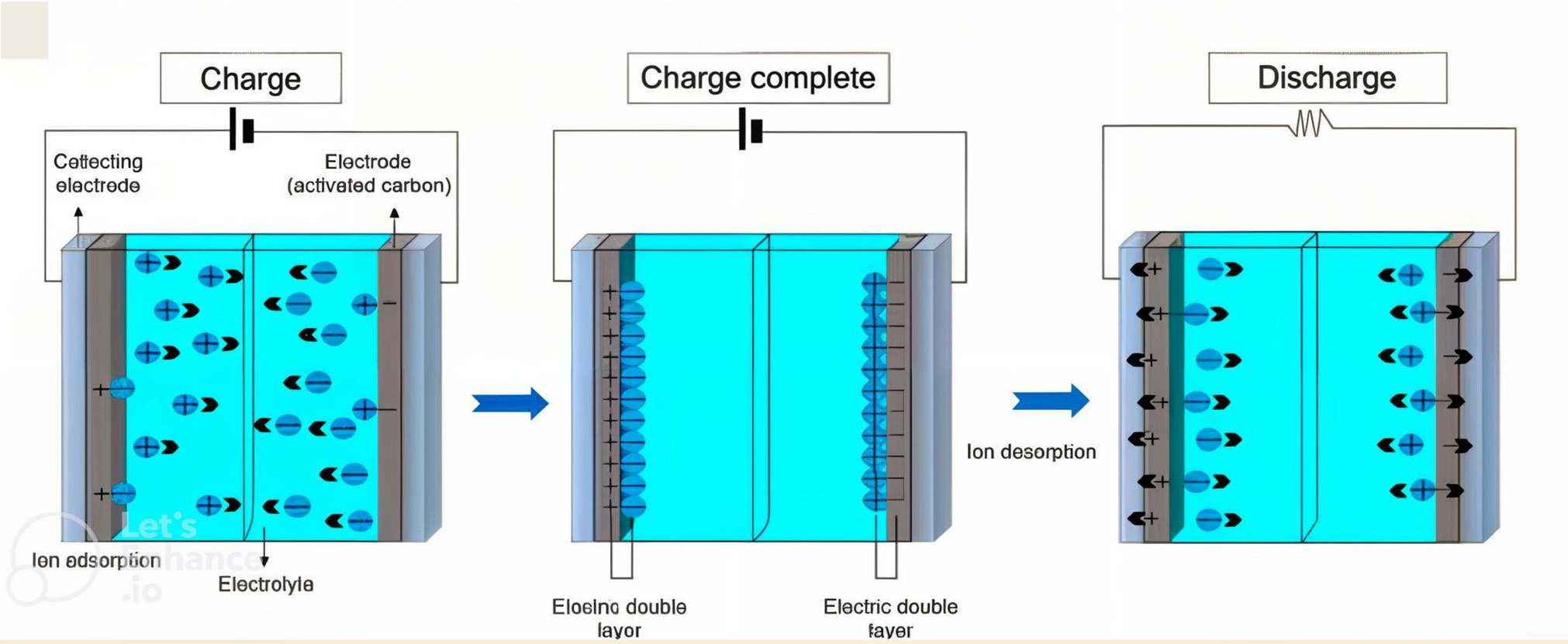


Figure 1. Specific Energy Vs Specific Power plot for various energy storage devices.

Working Mechanism of Supercapacitor :

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Applications of Supercapacitor :

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- ✓Energy Storage Device
- ✓Very high Charging and Discharging Application
- ✓Hybrid Electrical Vehicle



https://www.technologyreview.com/2009/10/19/209306/next-stop-ultracapacitor-buses/



Fig. Electric bus powered by Li-ion battery with support of supercapacitor

Literature Review

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Author	Electrode Material and Electrolyte	Investigation	Research Gap/Limitations
Khan et.al 2024	Crumple Graphene + NaCl	Crumpling Stage and Porosity of Graphene Electrode	Temp. variation and Use of Different electrolyte and Change of electrolyte concentration
Zhang et al. (2023)	Crumpled & porous graphene + Ionic liquid	Optimal porosity and crumpling degree for maximum capacitance	No aqueous electrolyte
Chen et al. (2022)	Crumpled graphene + Aqueous NaCl	Reported higher ion trapping and increased specific capacitance vs flat graphene	Electrolyte limited to NaCl
Qiao et al. (2016)	Graphene electrodes + KCl aqueous solution	Observed asymmetric ion distribution and solvent structuring near charged graphene	Only flat graphene considered

Objectives:

1. Investigate the effect of temperature variation on capacitance and ion distribution at the electrode in crumpled graphene supercapacitors.

2. Investigate the effect of different electrolytes at varying concentrations on capacitance and ion distribution at the electrode in crumpled graphene supercapacitors.

3. Investigate the effect of gas adsorption (Ar, N_2) on charge distribution at the electrode in crumpled graphene supercapacitors.

Previous Ref. paper Simulation And Result :

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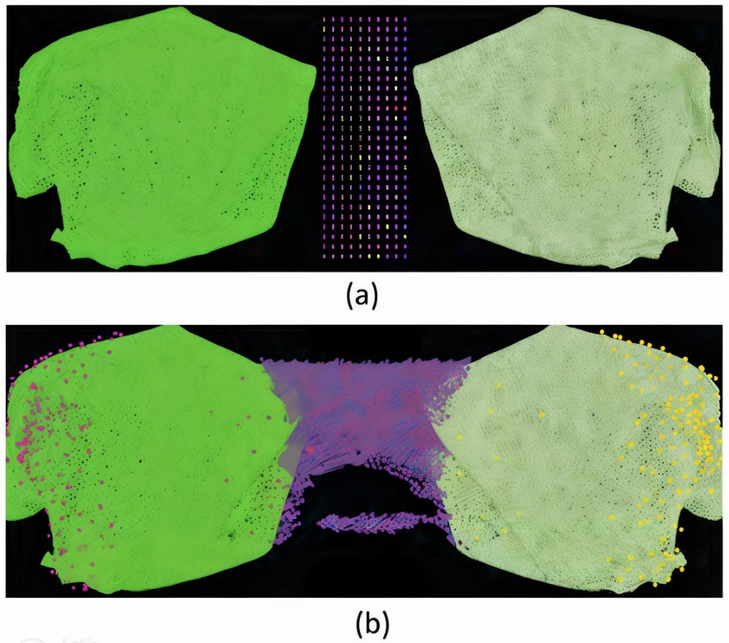


Fig 3. 2: EDLC scenarios (a) discharged state (b) charged state.

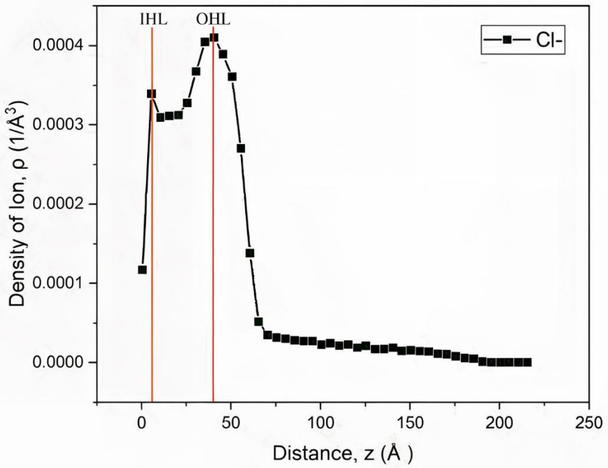


Fig 3. 3: Density profile of Cl^- for validated crumpled graphene model.

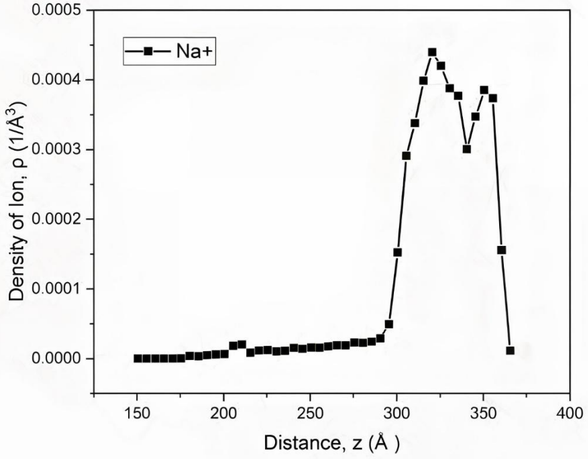


Fig 3. 4: Variation of Na^+ ion with for different spatial position

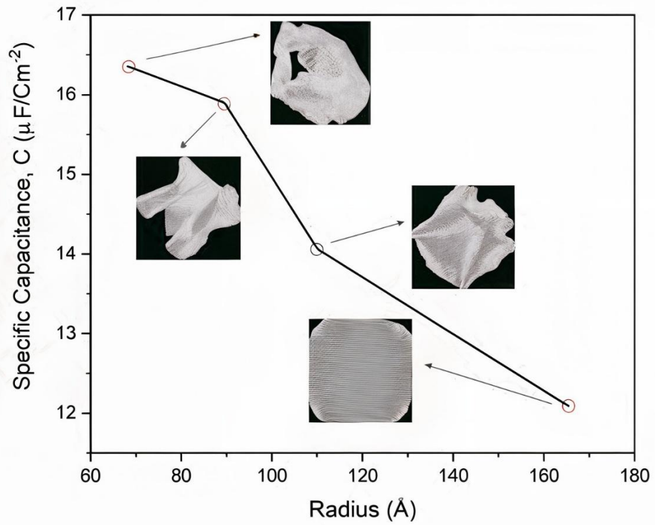


Fig 3. 11: Distribution of specific capacitance with varied crumpled radius

Capacitance Calculation:

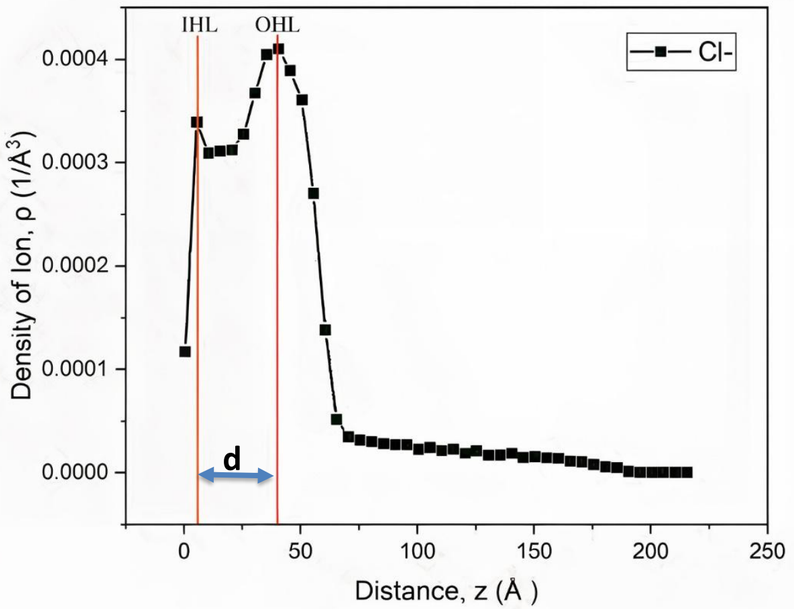


Fig 3. 3: Density profile of Cl^- for validated crumpled graphene model.

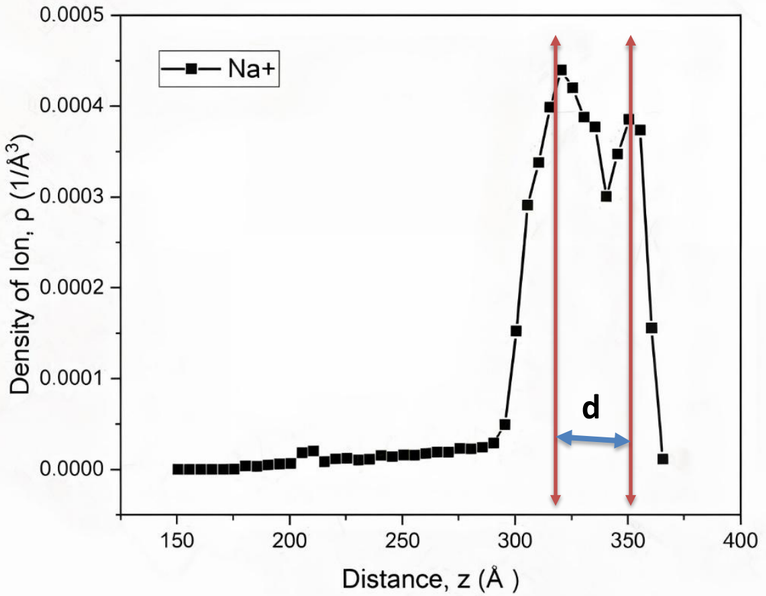


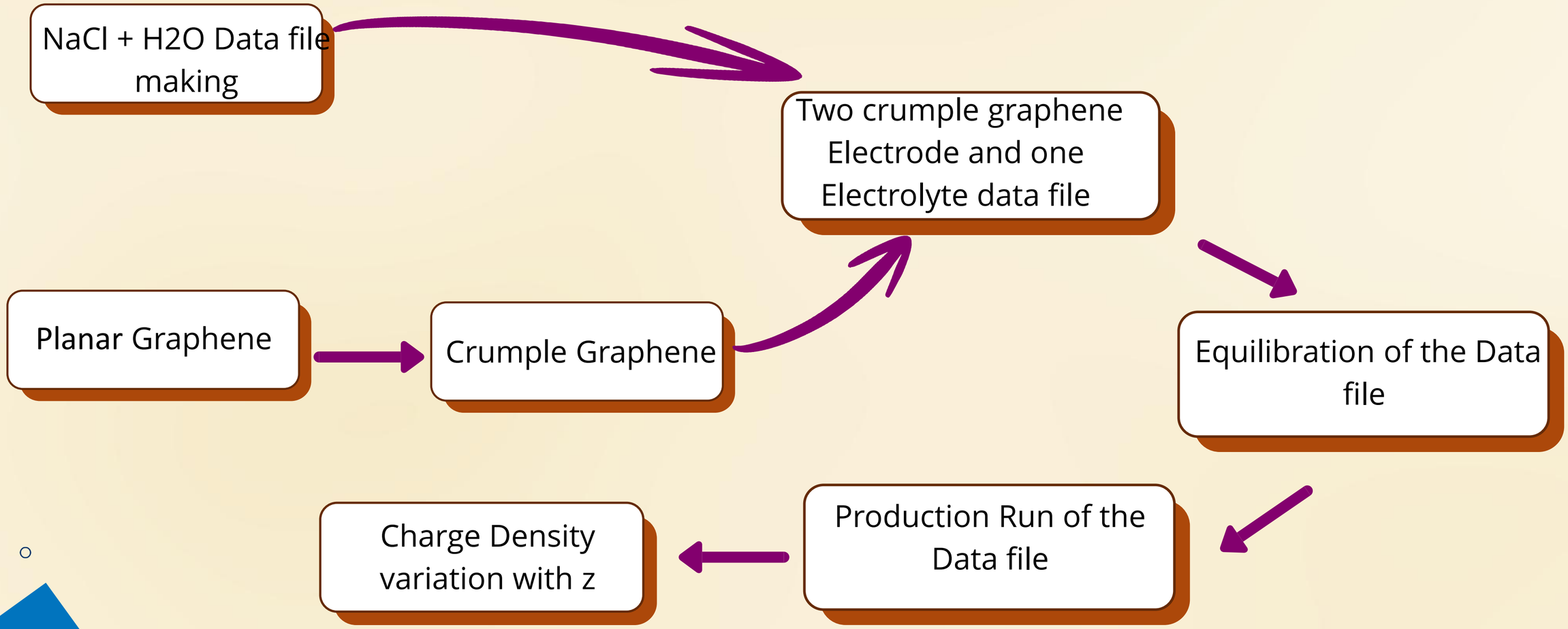
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$$C = \frac{\varepsilon_r \varepsilon_0 A}{d}$$

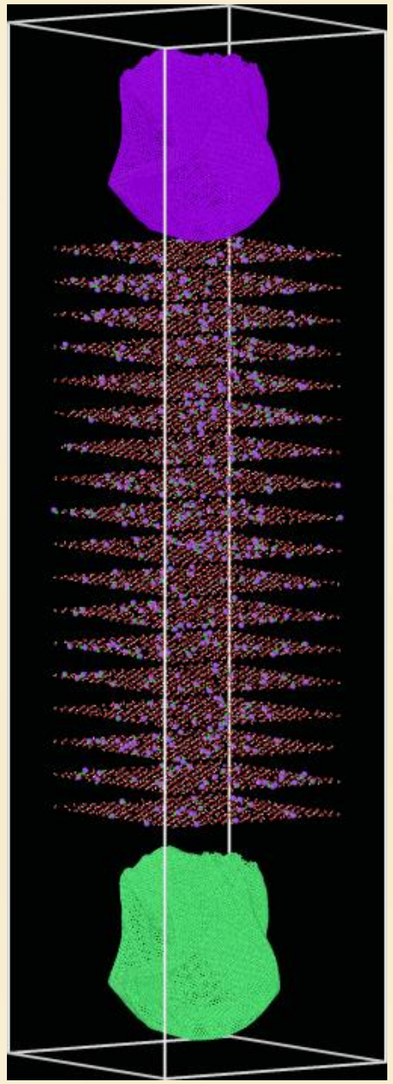
$$\frac{1}{C_T} = \frac{1}{C_A} + \frac{1}{C_C}$$

$$C = \frac{\varepsilon_r \varepsilon_0 A}{d}$$

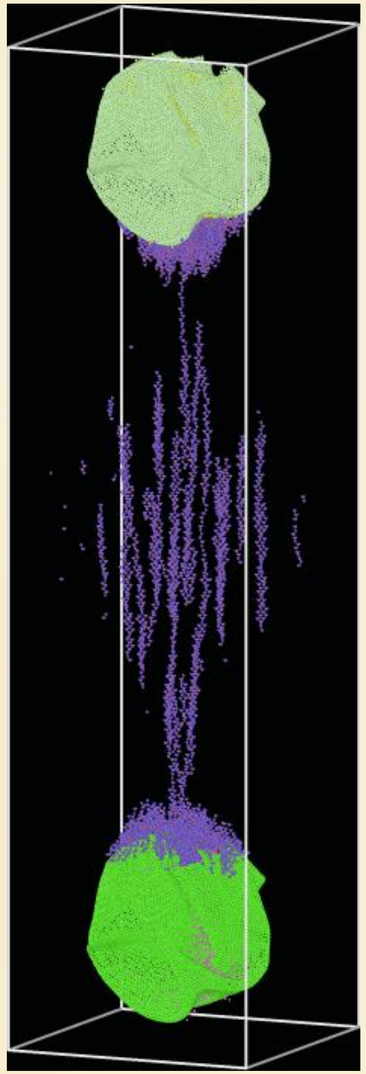
Simulation Methodology:



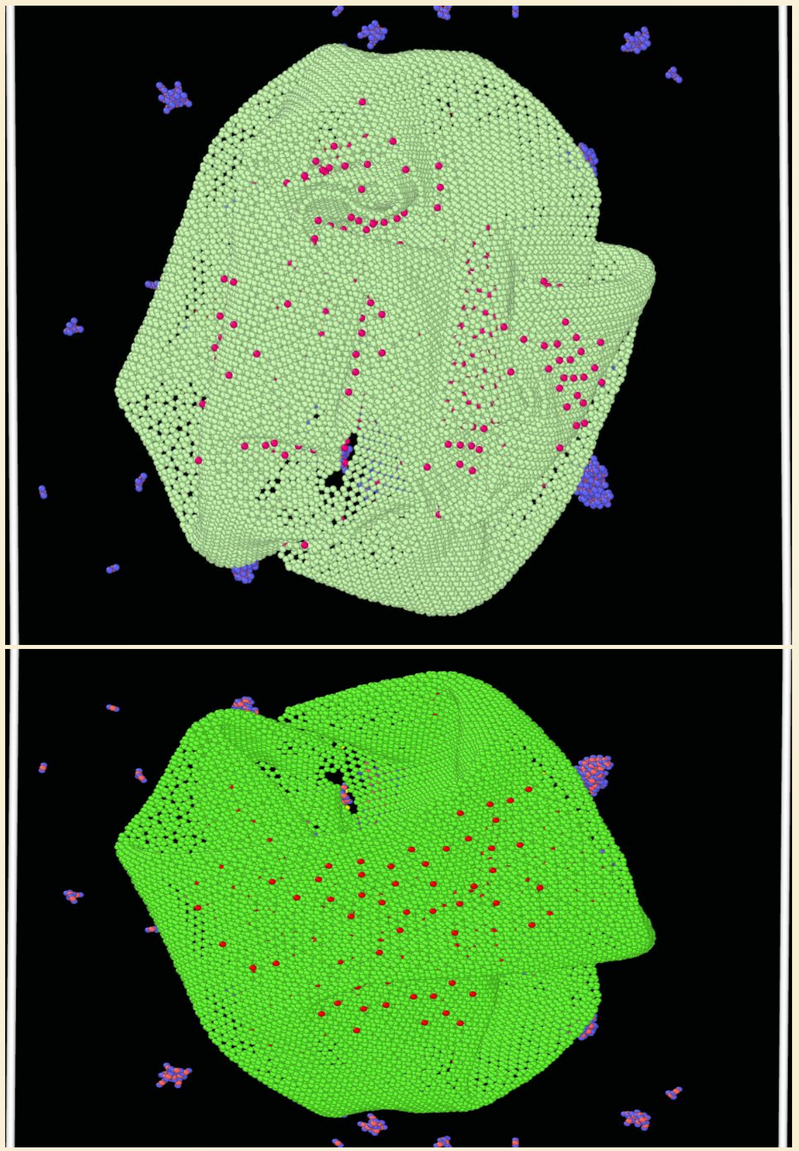
Validation Simulation result



Initial Data File

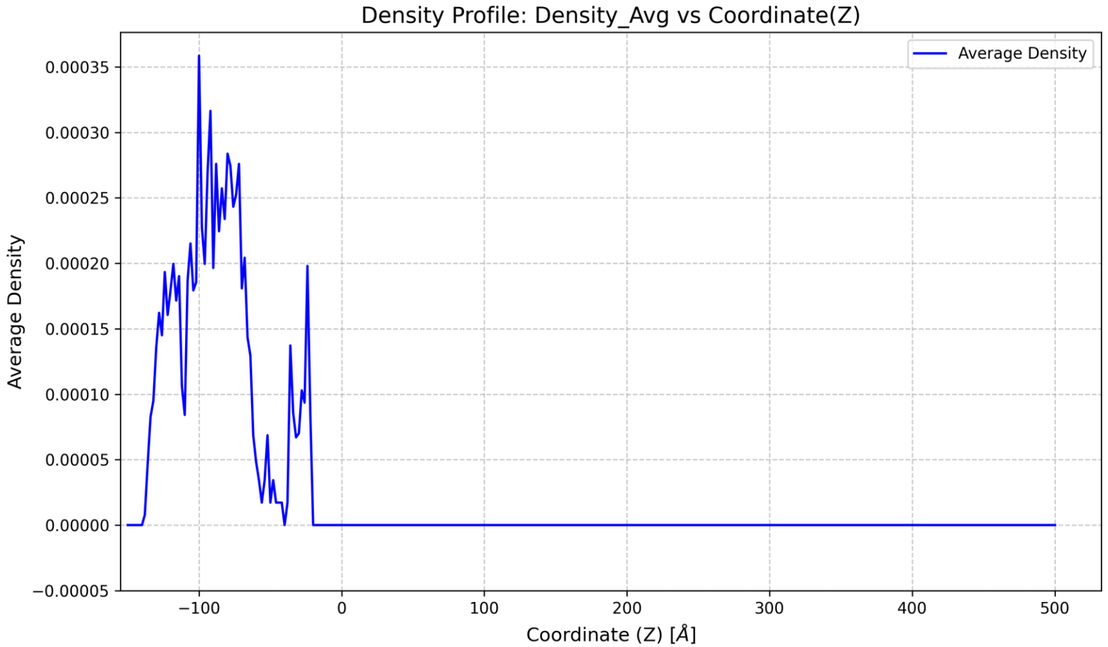


End of the Simulation

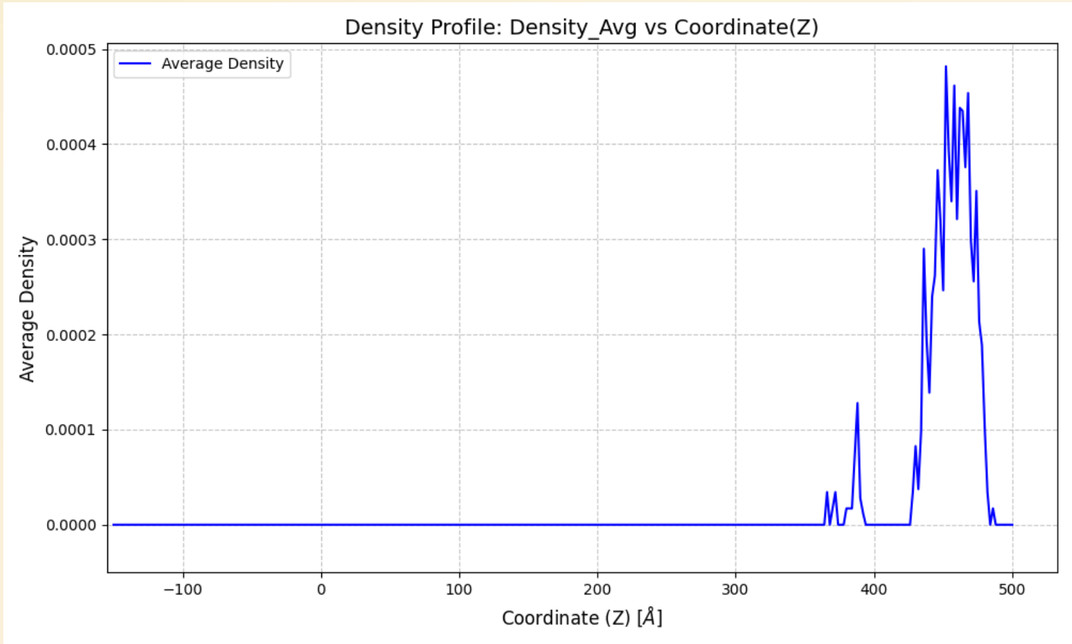


Validation Simulation result

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Cl Ion Density Variation with Position Z



Na Ion Density Variation with Position Z

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THANK YOU